

LandslideNEI — Master Technical Walkthrough Report

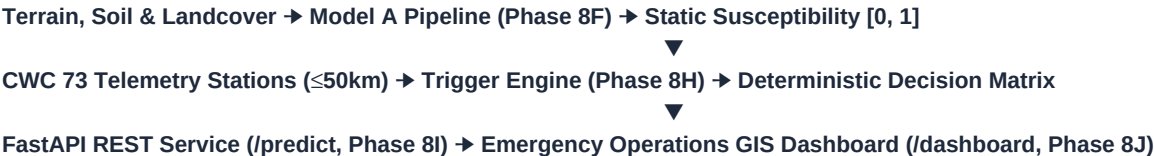
Comprehensive Architectural, Scientific, and Engineering Documentation (Phases 8F – 8J)

Executive Summary:

This master document encapsulates the end-to-end engineering progression of the LandslideNEI system for Smart India Hackathon (SIH 2024). Beginning with the static spatial Landslide Susceptibility Model (Phase 8F), location profiling (Phase 8G), dynamic rainfall telemetry fusion (Phase 8H), production REST API contract (Phase 8I), and interactive Emergency Operations Center GIS dashboard (Phase 8J).

Phase	Scope & Milestone	Key Scientific Artifacts	Status
Phase 8F	Static Landslide Susceptibility Model	Model A Pipeline, Spatial Block CV (5 folds, 41 tiles)	FROZEN
Phase 8G	Static Location Profiling Layer	LocationProfiler, Multi-raster feature extractor	FROZEN
Phase 8H	Dynamic Rainfall & Risk Fusion	CWC Telemetry (50km cap, 6h SLA), Decision Matrix	FROZEN
Phase 8I	Unified Risk API Contract	FastAPI REST Service, Pydantic V2 schemas	FROZEN
Phase 8J	Operational GIS Dashboard	Leaflet.js EOC UI, Demo Presets, Bilingual i18n	COMPLETE

System Architecture Flow



Core Invariants & Verification Status

- **Test Suite Status:** 151 / 151 Unit & Integration Tests Passing (100% Green).
- **Data Asset Integrity:** 10 / 10 Core Baseline Datasets & Model Artifacts Bitwise Intact.
- **Scientific Guardrail:** Operational risk levels are deterministic discrete decisions; fusion score is an engineering ordering synthesis, not an event probability.
- **Telemetry Guardrail:** Unobserved rainfall remains explicit null (never 0.0 mm); stations > 50 km discarded; telemetry > 6h marked STALE.
- **Interface:** Fully functional English & Tamil bilingual command center interface.